

Scourgify: Deterministic Spatial PDF Fixture

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Abstract

Scourgify is a local-first spatial canvas engineered for rigorous scientific document reading, bidirectional card linking, and offline annotation. This deterministic multi-page fixture guarantees reproducible ingestion, robust text coordinate extraction, and structured tabular validation.

1. Introduction and Architectural Motivation

Scientific inquiry demands seamless cross-referencing between hypotheses, mathematical derivations, and tabular empirical evidence. Traditional linear PDF viewers constrain cognitive synthesis. Scourgify treats document pages as spatially addressable artifacts on an infinite vector canvas. Users excerpt verbatim passages into atomic cards, cluster findings, and preserve provenance links.

2. Mathematical Formulations & Coordinate Transformations

Spatial projections map document space P into canvas coordinates C under scale zoom z :

Equation (1): Spatial Affine Projection

$$C(p, z, t) = z * (p - c_origin) + c_origin + [t_x , t_y]^T$$

where p is PDF coordinate, z is scale, and t is pan offset.

Equation (2): Variational Evidence Bound (ELBO)

$$L(\theta, \phi; x) = E_{\{q_{\phi}\}}[\log p_{\theta}(x | z)] - D_{KL}(q_{\phi}(z | x) || p(z))$$

optimizing retrieval likelihood under deterministic latent space regularization.

Equation (3): Multi-Head Cross Attention

$$Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V$$

computing card relevance against document selection tokens.

Equation (4): Conservation Energy Boundary

$$\Delta E = \int_{0}^T grad f(x(t)) \cdot x_{dot}(t) dt \leq epsilon_tolerance$$

bounding energy dissipation during continuous viewport inertia animations.

3. Empirical Performance & Benchmark Results

Table 1: Execution metrics for deterministic document processing pipelines.

Component	Throughput	Latency (p95)	Memory	Status
PDF Ingestion	142 pg/sec	6.8 ms	32.4 MB	VERIFIED
Spatial Canvas	60 FPS	16.6 ms	48.1 MB	VERIFIED
Card Extraction	310 cards/s	3.2 ms	12.0 MB	VERIFIED
Coordinate Index	890 queries/s	1.1 ms	18.5 MB	VERIFIED
Cold Relaunch	N/A	24.8 ms	64.0 MB	VERIFIED

4. Concluding Verification Invariants

The evaluation confirms that deterministic fixture synthesis fulfills all structural preconditions for automated testing: fixed bounding dimensions, invariant text baseline positioning, and rigorous metadata provenance.